

Jesús Ramírez Nava

Senior Software Engineer · Monterrey, Mexico

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Senior software engineer with 10 years in C/C++ — from 16-bit microcontrollers and CAN bus tooling at John Deere to Amazon Fire TV middleware and API backends today. I've led teams of 10+ engineers, served as Scrum Master, and designed the simulators, test tooling, and backends other teams depend on. Independently, I build web and AI products for clients — and real-time apps that make gatherings and events run smoother. Based in Monterrey (UTC-6) and open to remote work with US teams.

EXPERIENCE

Independent Software Engineer · [jesusrtech](#)

Jun 2026 – Present

Project: carterapp — offline-first personal finance app (carterapp.com.mx)

- _ Designed and built carterapp, a personal finance PWA for Mexico: log an expense in three taps, in Spanish or English, with MXN and USD kept side by side.
- _ Designed an offline-first sync model — records are written to the device first and synced with device-minted IDs, so entry never waits for the network and sync is idempotent.
- _ Enforced data integrity in the database itself: composite foreign keys keep every transaction in its account's currency and its category's type.
- _ Built monthly budgets, a card cut-off and payment calendar, a month-end balance projection, and the backend for subscriptions and push notifications.

React TypeScript PWA IndexedDB (Dexie) Supabase PostgreSQL Stripe Vercel

Independent Software Engineer · [jesusrtech](#)

Jun 2026 – Present

Project: Mi 70-20-10 — leadership coaching platform for eQual Business Solutions

- _ Designed and built a coaching platform where participants, coaches, and admins run a leadership-development program end to end.
- _ Secured the data with Postgres row-level security per role, so each user sees exactly what their role allows and roles can't be changed from the client.
- _ Built AI-assisted rubric evaluation of submissions, an AI coach chat, and retrieval over the program's material, with coaches approving what is released.
- _ Extracted a reusable React component library and automated deploys for the app and its backend functions.

React TypeScript Supabase PostgreSQL (RLS) Edge Functions Claude API
Gemini API Vitest Vercel

Senior Software Engineer · [Svitla Systems](#)

Mar 2026 – Present

Project: API backend abstraction to migrate a legal reporting system to a newer platform version

- _ Designing C++ features for an API backend abstraction layer that lets a legal reporting system migrate to its new platform version without breaking existing clients.
- _ Defining clean interfaces between the legacy C++ core and the new API so features can move over incrementally.
- _ Implementing new features and fixing defects through root cause analysis, working with beta users and QA.
- _ Integrating AI-assisted features into the backend and tuning them for response time and cost.

C++ API design Software architecture React AI integration

Senior Software Engineer · GlobalLogic

Aug 2023 – Dec 2025

Project: Amazon Fire TV next generation — porting a third-party video streaming app to new devices and Amazon's new non-Android OS

- _ Ported a major third-party video streaming app to Amazon's next-generation Fire TV devices, working in the middleware layer.
- _ Adapted Google's Cobalt application runtime to the new device hardware and to Amazon's new operating system, which is not based on Android.
- _ Implemented features and fixed defects through root cause analysis, working with beta users and test teams.
- _ Coordinated with partner engineering teams to route and escalate issues outside the project's scope.

C++ Google Cobalt Middleware API/ABI integration Third-party SDK integration Jira

Senior Software Engineer · GlobalLogic

Aug 2023 – Dec 2025

Project: Test environment to automate embedded-system release processes for an engineering-services provider

- _ Refactored a key desktop application with Python and PySimpleGUI, significantly improving readability and maintainability.
- _ Strengthened the architecture by applying SOLID principles and the listener pattern, producing a more robust, understandable codebase.
- _ Designed and built a project-manager feature from scratch to streamline creating and editing test profiles.

Python PySimpleGUI Git C++ Visual C++ Jira

Software Tech Lead · 3Pillar Global

Apr 2022 – Aug 2023

- _ Analyzed and fixed defects in existing features written in C++ and WinCC OA.
- _ Developed new functionality and unit tests, including keeping system status in sync with the data-queue handshake.
- _ Designed and implemented DevOps infrastructure to support a new licensing model, enabling migration to a modern tech stack.
- _ Built from scratch a Python 3 proof-of-concept API wrapper over crypto.com to simplify requests from higher abstraction layers — the bid was accepted.
- _ Levelled up in Go, Python, API design, and systems design through company training.

C++ WinCC OA Go Python AWS Docker CI/CD Linux Git

Software Supervisor / Technical Lead · John Deere

Feb 2021 – Mar 2022

- _ Led the Mexico team as tech lead and developer, fixing bugs in C++ engine-testing applications.
- _ Supervised 10+ engineers across 5 clients, guiding their professional and personal career paths.
- _ Interviewed candidates and designed team processes to onboard new engineers during headcount growth.

C C++ Qt RTOS Python CI/CD Linux Git

Independent Software Engineer · jesurtech

2020 – 2021

Project: Antártico — order management for cold-storage spaces

- _ Designed and built a web app to manage orders for cold-storage spaces, from the data model to deployment.
- _ Built it on Laravel and SQL behind a login, and deployed and ran it on a DigitalOcean server.

Laravel PHP SQL DigitalOcean Nginx

Senior Embedded Software Engineer · John Deere

Dec 2017 – Jan 2021

- _ Scrum Master and developer for embedded applications on 16-bit microcontrollers (C) and their GUIs (Qt C++).
- _ Led the Mexico Qt C++ team troubleshooting the tractor UI — from microcontroller buffer logic to data interpretation and on-screen visualization.
- _ Stood up and led a new team building simulators for the 16-bit microcontroller, the CAN bus, and virtual product interactions in C and C#.
- _ Led design and implementation of a virtual Ethernet simulation supporting Ethernet on 32-bit FreeRTOS microcontrollers.

_ Reported to users and stakeholders every 2-week sprint and presented the plan for each upcoming PSI.

C C++ C# Qt FreeRTOS CAN Ethernet Python Docker CI/CD Linux

Embedded Software Engineer · John Deere

Dec 2016 – Dec 2017

- _ Developed Qt UI applications, frontend and backend, in C++.
- _ Designed features to process data from a 16-bit microcontroller over CAN (J1939).
- _ Improved code quality by fixing inconsistencies and writing unit tests with Google Test.

C++ Qt CAN J1939 Google Test Jenkins Mercurial Docker Linux

Embedded Software Intern · John Deere

May 2016 – Dec 2016

- _ Designed and built from scratch a static code analyzer with a GUI that highlights bad coding practices, using Flex and Bison.
- _ Built electronic test boxes that emulate signals and communication protocols for device testing.

C C++ CAN Flex Bison HTML CSS

Computer Science Assistant · Tec de Monterrey

Jan 2015 – Aug 2015

- _ Wrote the base code for Kinect applications to improve the classroom experience through technology.
- _ Built Kinect interactions with hardware interfaces and controlled XAML apps through natural body movement in C#.

C# Kinect SDK XAML

EDUCATION

B.Sc. Digital Systems and Robotics Engineering

Tecnológico de Monterrey, Campus Monterrey

2016
MONTERREY, MEXICO

Technical Degree in Computer Science

ICE Computación

2011

Technical Degree in Computer Science / Business Administration

CBTis 67

2010

SKILLS

LANGUAGES

C C++ Python C# Go SQL PHP

EMBEDDED

16/32-bit microcontrollers FreeRTOS CAN / J1939 Ethernet Hardware simulation

FRAMEWORKS & UI

Qt React Laravel PySimpleGUI Google Test

BACKEND & SYSTEMS

API design API/ABI integration Third-party SDKs AI backends Systems design

DEVOPS & TOOLS

Docker AWS Jenkins CI/CD Linux Git Jira

LEADERSHIP

Tech Lead Scrum Master Team of 10+ Hiring & onboarding

PROJECTS & AWARDS

2026

Built apps.jesusrtech.com: real-time party games and group tools (Impostor, Werewolf, Peticiones, Pomodoro) that players join from their phones by QR code — React, Firebase Realtime Database, PWA.

2026	Built minivcs, a minimal git-like version control system in C++17 and Python: a content-addressed object store (blobs, trees, commits) with SHA-256 implemented from scratch and init, commit, log, and checkout. Both versions share one on-disk format, so either can read the other's repositories.
2019 - 2024	Designed and built a prototype of a next-generation learning management system (LMS) with Laravel and SQL, hosted on DigitalOcean.
2016	Built a MISRA-C analyzer that checks C code against the safety-critical coding standard, using Flex and Bison for syntax and semantic analysis.
2015	Built an open-source Android app for programming a robot over Bluetooth with interactive, graphical instructions.
2015	Built a computer-vision system for a Parrot drone that recognized objects around it and responded with matching flight routines.
2015	Parallelized a file-decryption program in C to speed it up.
2015	Created Robython, a high-level programming language and translator that lets children program hardware.
2014	Built a complete data-acquisition system for electrical conductivity: firmware in C plus a C# desktop app that reads the microprocessor and shares data over FTP.
2014	Built an IoT platform that monitors biosignals in real time with humidity, temperature, and heart-rate sensors. Winner, Intel challenge - SISCTI 40
2010	Designed an automatic, programmable pet-food dispenser in C. State winner, DGETI prototypes competition - national finalist

LANGUAGES

Spanish — Native English — B2

CERTIFICATIONS

Generative AI Leader	GOOGLE CLOUD · AUG 2025
Google Cybersecurity Specialization	GOOGLE · DEC 2023